EpiBench: Verifiable Evaluation of AI Agents on Epigenomics Analysis

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ABSTRACT

We introduce EpiBench, a verifiable benchmark for short-horizon epigenomics analysis. EpiBench evaluates whether agents can make well-defined analysis decisions from realistic workflow states and return deterministically gradable answers. The benchmark includes 106 evaluations across CUT&Tag/CUT&RUN, ATAC-seq, ChIP-seq, and DNA methylation workflows. Across 5,088 valid trajectories from 16 model-harness pairs, no system passed a majority of attempts: GPT-5.5 / Pi leads at 45.0%, followed by GPT-5.5 / OpenAI Codex at 39.9%. Claude Opus 4.8 Max / Pi and GPT-5.4 / Pi each achieved 39.0%. Performance varies across assay types, and many failed runs still contain parts of the correct answer. Agents often found the right files and computed useful intermediate results, but failed when the task required deeper, assay-specific scientific judgment.

Topline Benchmark Performance

We ran the benchmark across 16 model-harness pairs. Pi denotes the Pi terminal coding harness. Passing requires exact recovery of the graded structured answer for an evaluation attempt. The result set contains 5,088 valid trajectories in a balanced three-replicate design: 106 evaluations for each of 16 model-harness pairs.

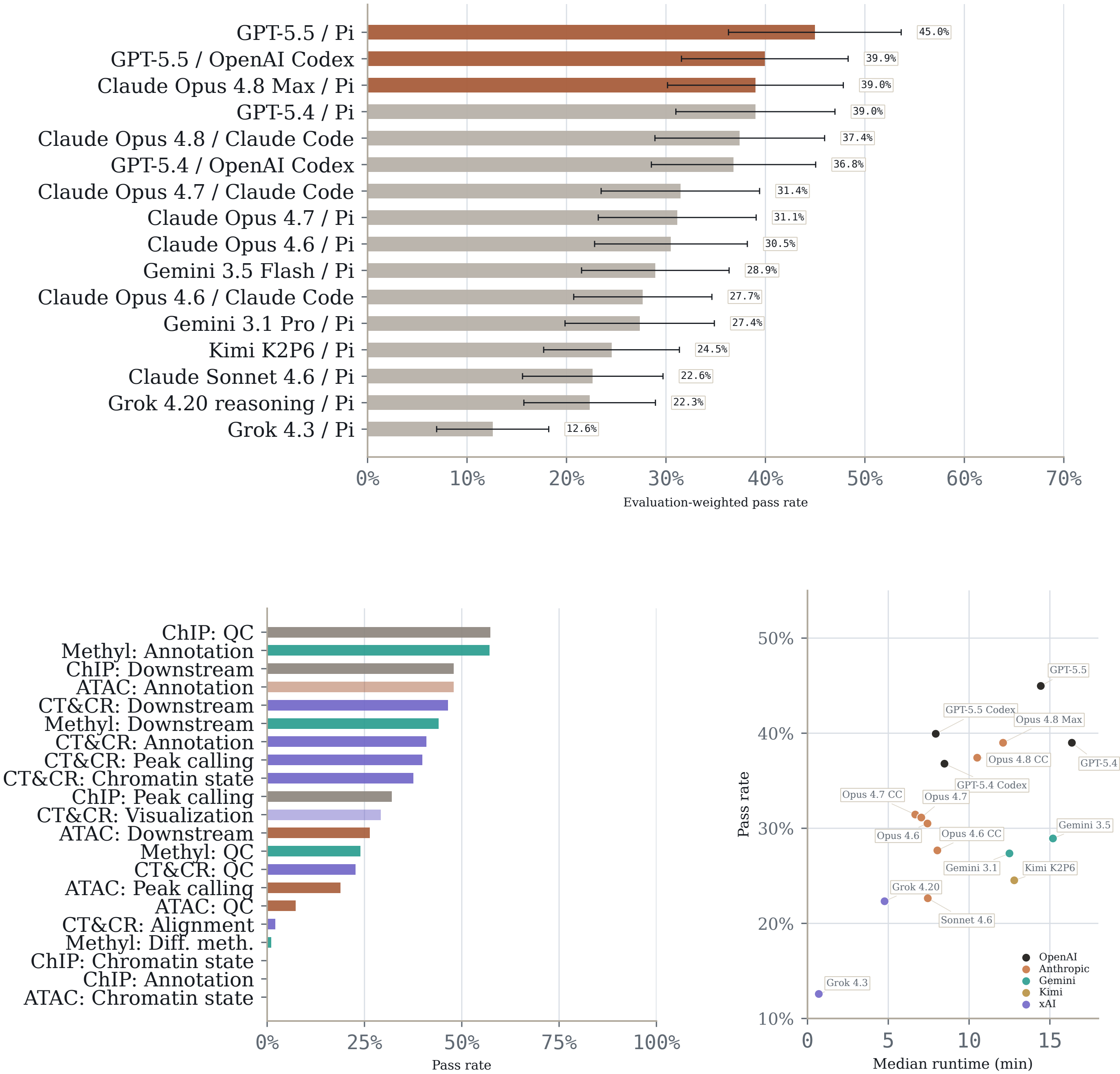

Figure 1: Topline benchmark performance. Top: evaluation-weighted pass rate by model-harness pair across all trajectories. The top panel shows Student t confidence intervals over per-evaluation replicate means. Bottom left: pass rate by assay and task family. Bottom right: pass rate against median runtime. Evaluation-level replicate statistics are reported in Table 2.

Introduction

Epigenomic assays measure the regulatory state of the genome: chromatin accessibility, histone modifications, DNA methylation, and other molecular signals that link genotype to cell state and gene regulation [1, 2]. But raw epigenomics data are not directly interpretable. Drawing scientific conclusions requires a sequence of analysis decisions across read alignment, peak calling, motif interpretation, sample aggregation, normalization, and genomic annotation. Small procedural choices can change the final biological conclusion. AI agents show early promise in human-guided analysis of biological data. They can inspect files, write code, invoke command-line tools, and exercise some scientific judgment in analysis tasks. But they remain prone to hallucination and unreliable biological judgment, especially across diverse tissues, diseases, study designs, and assay types.

Existing biology benchmarks increasingly test practical analysis in other data modalities [6, 7], but, to our knowledge, none focus on epigenomics analysis alone. We introduce EpiBench, a verifiable benchmark for short-horizon epigenomics tasks. Each evaluation snapshots a realistic workflow state immediately before a target result. The agent receives relevant files, metadata, and task context, then must interact with the data and return a structured answer graded deterministically. The benchmark includes 106 evaluations across CUT&Tag/CUT&RUN, ATAC-seq, ChIP-seq, and DNA methylation workflows, covering quality control, peak calling, chromatin-state analysis, genomic annotation, differential methylation, downstream quantitative analysis, alignment, and visualization.

Across 5,088 valid trajectories from 16 model-harness pairs, no system passed a majority of attempts. GPT-5.5 / Pi leads at 45.0%, followed by GPT-5.5 / OpenAI Codex at 39.9%. Claude Opus 4.8 Max / Pi and GPT-5.4 / Pi each achieved 39.0%. The failures show that current agents can sometimes make reasonable progress with epigenomics data, but still struggle with some analysis decisions, especially those that require biological reasoning: choosing which samples to compare, accurate normalization, identifying appropriate genomic features, and reporting appropriate statistics.

Benchmark Construction

To construct a benchmark that approximates real epigenomics analysis, we derived evaluations from practical workflows across CUT&Tag/CUT&RUN, ATAC-seq, ChIP-seq, and DNA methylation. We decomposed analysis workflows into short, gradeable decisions: quality control, peak calling, chromatin-state comparisons, genomic annotations, methylation statistics, alignment checks, and downstream quantitative analyses such as motif, gene-set, regulatory-domain, and cross-modality summaries. At each step, we sought to isolate the empirical decision that determines the result and formalize it as a deterministically gradable answer. The final evaluation suite consists of 106 problems spanning four assay families and eight task categories. Each problem includes a snapshot of real assay outputs immediately before a target analysis step, metadata needed to interpret the files, a high-level task description, and a deterministic grader that evaluates whether the agent recovered the requested empirical result.

We follow three design criteria. First, tasks must be verifiable: each evaluation admits an automatically checkable success condition, such as a numerical interval check, structured label match, or all-of field comparison. Second, tasks should be scientifically durable: the intended answer should be recoverable across reasonable analysis choices, so the benchmark measures epigenomics reasoning rather than incidental implementation details. Third, tasks should resist shortcuts: agents must interact with the provided data artifacts, and we remove items that can be solved reliably by trivial heuristics or prior biological knowledge alone.

Tasks are designed to specify what should be recovered rather than how it should be recovered. Some procedural evaluations name a specific operation, but most scientific evaluations leave the agent to choose reasonable analysis methods to accomplish a goal. A correct answer should therefore reflect the result supported by the specific assay context, not a nearby generic workflow convention or a plausible biological expectation.

All evaluations underwent manual quality control. We inspected agent trajectories across runs to identify prompts that over-specified the method, answers that could be obtained through shortcuts, and graders that failed to distinguish supported results from plausible but unsupported biological outputs.

Evaluation Inventory

The benchmark covers four assay types and eight task categories. CUT&Tag/CUT&RUN tasks are drawn from zebrafish chromatin workflow snapshots and span chromatin state analysis, QC, peak calling, downstream analysis, genomic annotation, alignment, and visualization. ATAC-seq tasks use pediatric B-ALL ATAC/RNA resources from GSE161501 and related integrative workflow artifacts [3], with downstream analysis dominating. ChIP-seq tasks use B-ALL H3K27ac resources from GSE211631 [4] and include peak calling, chromatin state analysis, QC, genomic annotation, and downstream analysis.

Methylation tasks are derived from the ESCC WGBS/RNA resources GSE149608 and GSE149609 [5], covering QC, downstream analysis, genomic annotation, and differential methylation.

Across all 106 evaluations, 34 are downstream analysis tasks, 22 are QC, 16 are chromatin state analysis, 14 are peak calling, 12 are genomic annotation, 4 are differential methylation, 3 are alignment, and 1 is visualization. The result set contains 5,088 valid trajectories from 16 model-harness pairs, with three attempts for each evaluation and model-harness pair.

Assay type	Evals	Sources	Task categories
CUT&Tag/CUT&RUN	47	Zebrafish chromatin workflow snapshots	Chromatin state (12), QC (10), peak calling (9), downstream analysis (7), genomic annotation (5), alignment (3), visualization (1)
ATAC-seq	24	B-ALL ATAC/RNA (GSE161501 and integrative workflow artifacts)	Downstream analysis (17), chromatin state (2), peak calling (2), QC (2), genomic annotation (1)
ChIP-seq	10	B-ALL H3K27ac (GSE211631)	Peak calling (3), chromatin state (2), QC (2), downstream analysis (2), genomic annotation (1)
Methylation-seq	25	GSE149608, GSE149609	QC (8), downstream analysis (8), genomic annotation (5), differential methylation (4)

Table 1: Evaluation inventory. The benchmark groups evaluations by assay type while varying the source workflow, task category, and expected answer surface within each group.

Results

No agent reaches a 50% endpoint pass rate

Across 16 model-harness pairs and 5,088 valid trajectories, deterministic endpoint grading produced 1,578 passes (31.0%). No observed endpoint pass rate reached 50%. GPT-5.5 / Pi was the descriptive leader at 45.0% (95% confidence interval (CI), 36.3–53.7), followed by GPT-5.5 / OpenAI Codex at 39.9% (31.6–48.3). Claude Opus 4.8 Max / Pi and GPT-5.4 / Pi each passed 39.0%.

The leading systems form a broad tier rather than a stable ordering: their confidence intervals overlap, and their pass rates are close. Even the strongest model-harness pair passed fewer than half of endpoint-graded attempts, indicating that short-horizon epigenomics analysis remains difficult for current agents.

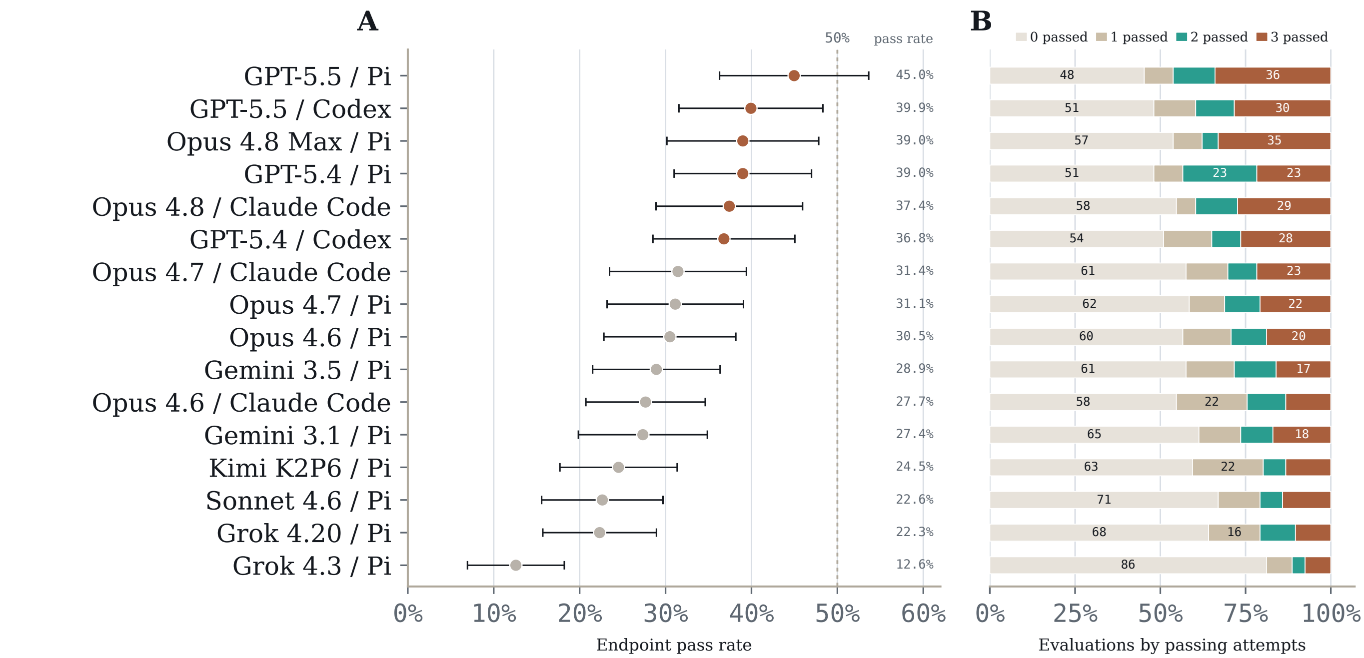

Figure 2: Endpoint pass rates across model-harness pairs. Panel A: endpoint pass rate across all 5,088 trajectories, with 95% confidence intervals and a 50% reference line. Panel B: evaluation-level replicate outcomes over the same model order, partitioning the 106 evaluations for each model-harness pair by whether zero, one, two, or all three attempts passed.

Model / harness	Pass %	95% CI	≥1/3	≥2/3	3/3
GPT-5.5 / Pi	45.0	36.3–53.7	58/106	49/106	36/106
GPT-5.5 / OpenAI Codex	39.9	31.6–48.3	55/106	42/106	30/106
Claude Opus 4.8 Max / Pi	39.0	30.2–47.8	49/106	40/106	35/106
GPT-5.4 / Pi	39.0	31.0–47.0	55/106	46/106	23/106
Claude Opus 4.8 / Claude Code	37.4	28.9–46.0	48/106	42/106	29/106
GPT-5.4 / OpenAI Codex	36.8	28.5–45.1	52/106	37/106	28/106
Claude Opus 4.7 / Claude Code	31.4	23.5–39.4	45/106	32/106	23/106
Claude Opus 4.7 / Pi	31.1	23.2–39.1	44/106	33/106	22/106
Claude Opus 4.6 / Pi	30.5	22.8–38.2	46/106	31/106	20/106
Gemini 3.5 Flash / Pi	28.9	21.5–36.3	45/106	30/106	17/106
Claude Opus 4.6 / Claude Code	27.7	20.7–34.6	48/106	26/106	14/106
Gemini 3.1 Pro / Pi	27.4	19.9–34.9	41/106	28/106	18/106
Kimi K2P6 / Pi	24.5	17.7–31.4	43/106	21/106	14/106
Claude Sonnet 4.6 / Pi	22.6	15.6–29.7	35/106	22/106	15/106
Grok 4.20 reasoning / Pi	22.3	15.7–28.9	38/106	22/106	11/106
Grok 4.3 / Pi	12.6	6.9–18.2	20/106	12/106	8/106

Table 2: Endpoint pass rates by model-harness pair. Pass rates are computed over the balanced three-replicate result set. The 95% CI is computed over per-evaluation replicate means. The last three columns report the number of evaluations, out of 106, with at least one, at least two, or all three replicate attempts passing. Runtime and cost are treated as operational diagnostics rather than leaderboard columns.

Endpoint pass rates vary by assay type

Endpoint pass rates differed by assay type. CUT&Tag/CUT&RUN had the highest aggregate pass rate at 34.0%, followed by methylation-seq at 33.3% and ChIP-seq at 30.6%. ATAC-seq was lowest at 22.8%. Thus, the largest separation was between ATAC-seq and the other assay families, rather than between a single high-performing assay and the rest of the benchmark.

The assay groups differed in both size and task composition. CUT&Tag/CUT&RUN contributed 47 evaluations, methylation-seq 25, ATAC-seq 24, and ChIP-seq 10, with different proportions of QC, peak calling, annotation, chromatin-state, and downstream-analysis tasks. Because assay type, source workflow, and task mix are coupled in this benchmark, we treat assay-level pass rates as descriptive summaries rather than controlled estimates of intrinsic assay difficulty.

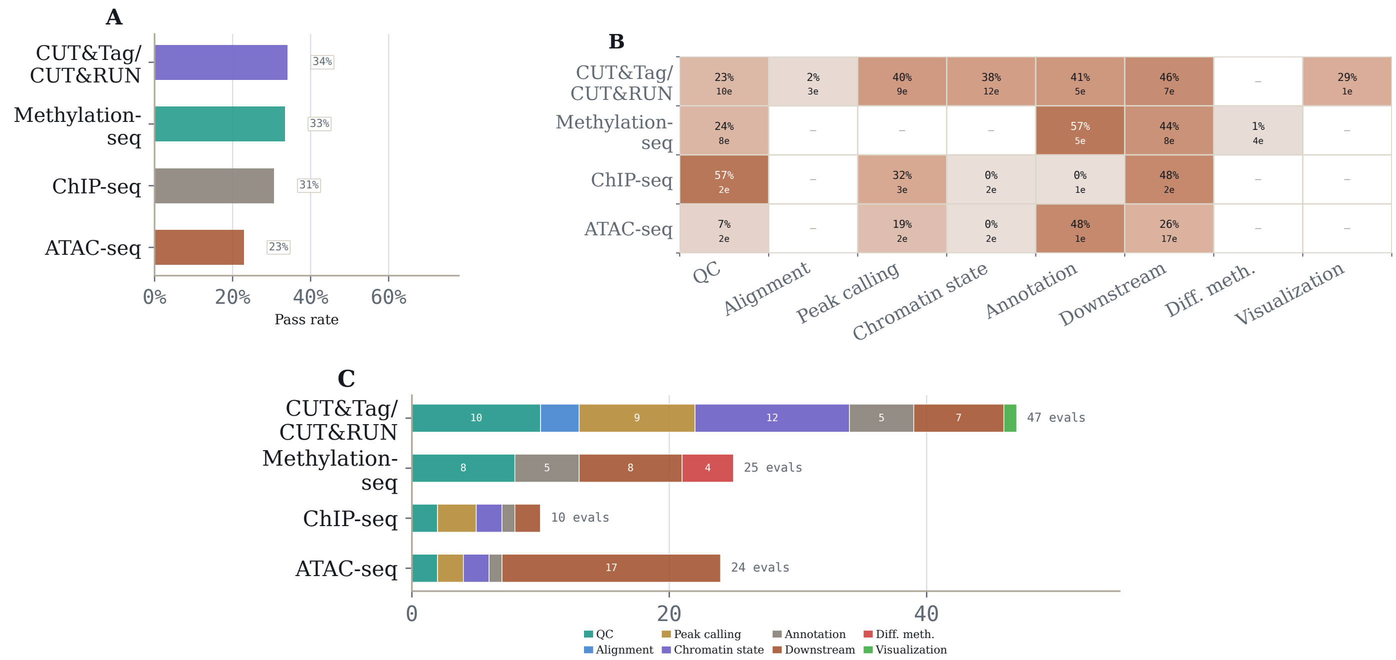

Figure 3: Performance varies by assay type. Panel A: pass rate by assay type. Panel B: assay-by-task-category pass-rate heatmap; cell text reports pass rate and the number of evaluations in that cell. Panel C: evaluation-count composition by assay type, showing the task mix behind each assay summary.

Many failures trace to plausible but wrong scientific choices

Endpoint failures clustered around decision points where a generic bioinformatics workflow diverged from the assay-specific evidence in the provided files. Manual review of 25 evaluations identified recurring errors in which statistic to rank, which threshold to apply, which biological unit should define the denominator, which file or feature namespace represented the signal, and when to trust file-derived evidence over familiar biological priors (Figure 4). These choices were often small in implementation terms, but they determined whether the final structured answer matched the data.

Component-level grading supported the same interpretation. Among 5,051 trajectories with field-level grader scores, individual required fields passed 68.2% of the time, compared with a 31.0% endpoint pass rate. The gap indicates that many failed endpoints contained some correct pieces, but missed the exact data layer, statistical choice, biological unit, or interpretation field required by the task.

Examples span the assay families. In CUT&RUN spike-in normalization, choosing Bowtie2 end-to-end alignment rather than local alignment reduced spike-in recovery and distorted downstream normalization. In WGBS outputs, treating the two Bismark rows for a CpG dinucleotide as independent sites changed coverage and differential-methylation statistics. In ATAC-seq, using paired-end BAMs directly with MACS3 `-f BAMPE` bypassed the intended single-base Tn5 insertion-site representation, shifting peak calls and downstream motif or integration results. In interpretation tasks, agents sometimes substituted a literature-derived mechanism for the comparison supported by the files.

A related pattern appeared when agents computed the correct result but did not submit it. Across reviewed GPT-5.5 trajectories for CUT&Tag/CUT&RUN and methylation-seq, this occurred in seven evaluations; in four of those, the correct answer was visible in the agent's own output before being replaced by a more familiar workflow default or biological expectation. These cases separate tool execution from scientific judgment: the result was available, but the agent selected the less supported answer.

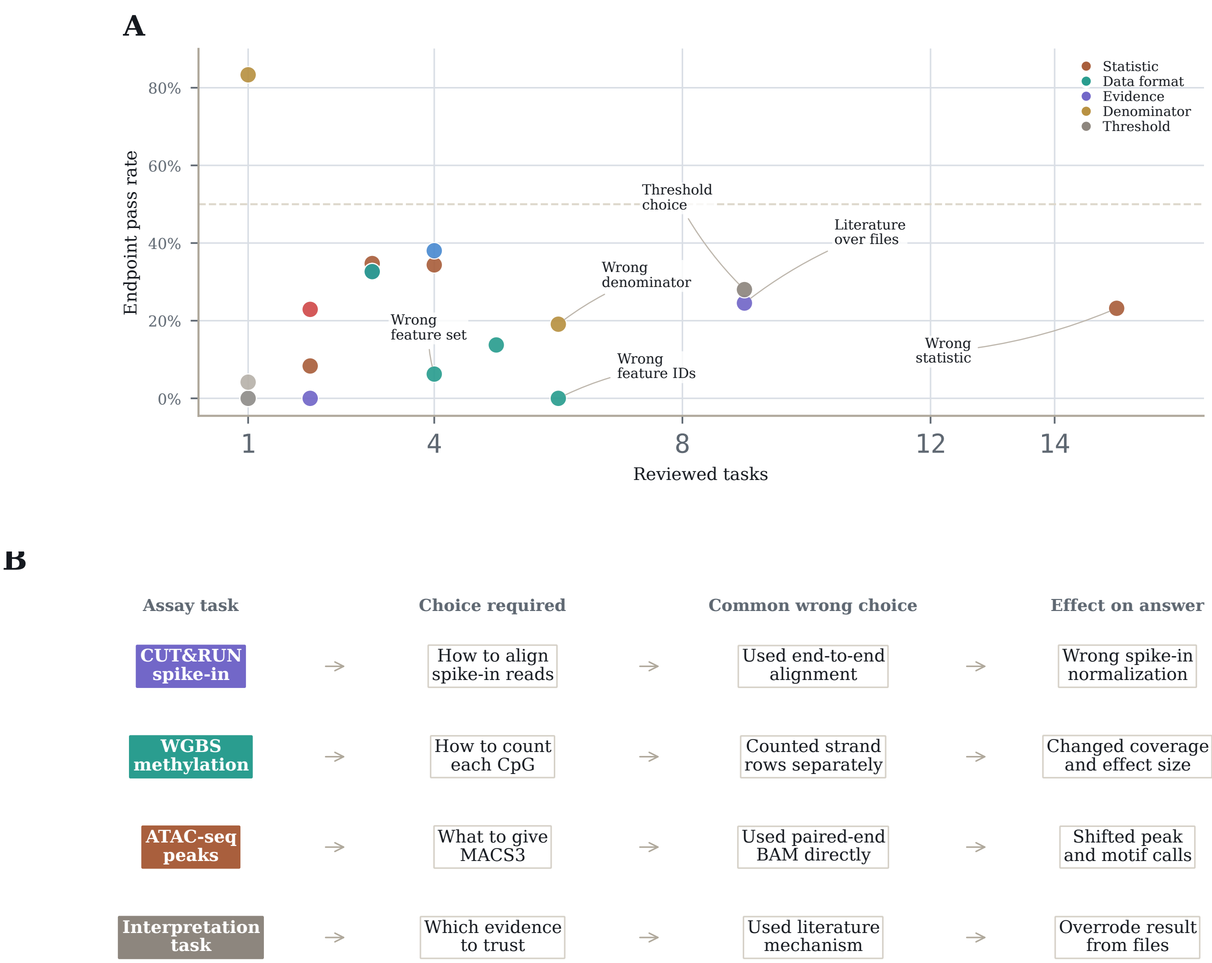

Figure 4: Many failures trace to plausible but wrong scientific choices. Panel A summarizes recurring labels from manual review. The x-axis counts reviewed evaluations carrying each label; lower y-values mark labels associated with lower endpoint pass rates. Panel B gives representative assay choices where a plausible default changed the endpoint answer. Manual review covers 25/106 evaluations and is diagnostic rather than exhaustive.

Discussion

EpiBench is designed around a practical question: can an agent recover a specific empirical result from epigenomics artifacts immediately upstream of that result? The tasks are short, but the decisions are not trivial. Assay representation, normalization state, genomic unit, comparator choice, and answer surface can determine whether the submitted answer matches the supported biological conclusion.

The results show that these local decisions remain unreliable. The strongest model-harness pair passed 45.0% of endpoint attempts, and many failures occurred despite partial progress. Field-level scores and trajectory review suggest that agents often found relevant files or computed useful intermediate values, but then selected a familiar workflow default, literature-derived mechanism, or nearby statistic that was not supported by the provided data. This separates tool use from scientific judgment: the agent can run the workflow and still submit the wrong biological result.

The benchmark has several limitations. The result set is balanced across model-harness pairs, but the task inventory is not balanced across assay families: CUT&Tag/CUT&RUN (47 evaluations) and methylation-seq (25) contribute more evaluations than ATAC-seq (24) or ChIP-seq (10), and downstream analysis plus QC tasks dominate the inventory. Some failure labels also recur across related evaluations, so aggregate scores partially reflect repeated tests of the same underlying decision. This mirrors real workflow failure modes, but it should temper per-evaluation independence claims. Finally, deterministic graders intentionally constrain the answer surface; they are useful for measurement, but they do not capture every scientifically valid analysis path. EpiBench therefore tests a defined sample of practical epigenomics skills, not general epigenomics reasoning in full.

Methods

Evaluation construction

Candidate evaluations were derived from real epigenomics analysis workflows. An evaluation was retained when the target result could be reproduced from the provided data, expressed through a constrained final answer, and graded deterministically. Tasks were revised or removed when prompts over-specified the method, when reasonable analysis choices produced incompatible answers, or when the grader could not distinguish a supported result from plausible but unsupported biological output.

Agent runs and aggregation

Each model-harness pair was run with up to three replicates per evaluation. All 5,088 trajectory outputs were downloaded and parsed successfully. The result set contains a complete three-replicate design for 106 evaluations and 16 model-harness pairs. Failed, malformed, or nonpassing endpoint grader outputs are counted as nonpassing attempts when present; no downloaded output was missing or invalid in this analysis. Component field scores were available for 5,051 trajectories and are analyzed only as diagnostics.

Statistical analysis

The primary metric is evaluation-weighted endpoint pass rate. For each model-harness pair and evaluation, we first compute the mean pass rate across the three replicates, then report the mean across evaluations. Because the result set is balanced, this estimate is numerically identical to the raw run-level pass rate. Confidence intervals use the Student t distribution over per-evaluation replicate means, matching the SpatialBench aggregation convention [6]. We also report evaluation-level replicate summaries: whether a model-harness pair passed any attempt, a majority of attempts, or all three attempts for an evaluation. Component field scores and failure-mode labels are reported as diagnostics rather than benchmark scores.

Data availability

Results files, public example evaluations, and representative trajectories will be made available at `https://github.com/latchbio/epibench`.

References

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